INCH-POUND

MIL-DTL-24174B(SH)
19 March 2026
SUPERSEDING
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2 October 1987

DETAIL SPECIFICATION

CONTROLS, NAVIGATION LIGHTS, GENERAL SPECIFICATION FOR

This specification is approved for use by Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for controls for flashing of navigation lights under conditions requiring sequence flashing (pulsation) of these lights for signaling purposes.

1.2 Classification. Controls are of the types specified on the individual specification sheet (see 3.1 and 6.2).

1.3 Part or identifying number (PIN). PINs to be used for navigation light controls acquired to this specification are created as follows:

M	**24174**	**/X**	**-**	**001**
Prefix for Military Specification	Specification Number	Spec Sheet Number	-	Type

Examples: M24174/3-001
M24174/4-001

2. APPLICABLE PUBLICATIONS

2.1 General. The publications listed in this section are specified in sections 3 and 4 of this specification. This section does not include publications cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, users of this document are cautioned that they must meet all specified requirements of publications cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government publications.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901 - Shock Tests, H.I. (High Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

MIL-DTL-917 - Electric Power Equipment, Basic Requirements for

MIL-DTL-2036 - Enclosures for Electric and Electronic Equipment, Naval Shipboard

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using ASSIST at https://assist.dla.mil.

AMSC N/A FSC 6110

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MIL-DTL-15024 - Plates and Tags for Identification of Equipment, General Specification for

MIL-P-15024/5 - Plates, Identification

MIL-T-16366 - Terminals, Electrical Lug and Conductor Splices, Crimp-Style

MIL-E-24142 - Enclosures for Electrical Fittings and Fixtures, General Specification for

MIL-DTL-24174/3 - Control (Blinker) Key, Navigation Lights Symbol No. 978.2

MIL-DTL-24174/4 - Controls, Pulsator, Submarine Identification Light, A.C. 125 Volts, Symbol No. 932

MIL-I-24768/1 - Insulation, Plastic, Laminated, Thermosetting, Glass-Cloth, Melamine-Resin (GME)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)

MIL-STD-202 - Electronic and Electrical Component Parts

MIL-STD-740-2 - Structureborne Vibratory Acceleration Measurements and Acceptance Criteria of Shipboard Equipment

MIL-STD-1474 - Noise Limits

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-454 - General Guidelines for Electronic Equipment

(Copies of these publications are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D5948 - Standard Specification for Molding Compounds, Thermosetting

(Copies of this publication are available online at www.astm.org.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B1.1 - Unified Inch Screw Threads (UN, UNR, and UNJ Thread Forms)

(Copies of this publication are available online at https://www.asme.org)

AMERICAN WELDING SOCIETY (AWS)

WHB-1.10 - Welding Handbook Volume 1 – Welding and Cutting Science and Technology

(Copies of this publication are available online at www.aws.org.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code

(Copies of this publication are available online at www.nfpa.org.)

SAE INTERNATIONAL (SAE)

SAE AS7928 - Terminals, Lug: Splices, Conductor: Crimp Style, Copper, General Specification for

(Copies of this publication are available online at www.sae.org.)

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.3 Materials. Materials shall be as specified (see 6.2). Materials not specified shall be, insofar as practicable, of a type, class, form, and grade that is readily available from normal sources of supply.

3.3.1 Prohibited material. Any material identified as prohibited material in accordance with MIL-DTL-917 shall not be used.

3.3.2 Insulating materials.

3.3.2.1 Molded plastics. Molded insulated parts shall be in accordance with type GPI-20, MAI-60, or MMI-30 of ASTM D5948.

3.3.2.2 Laminated plastics. Laminated plastic material shall be in accordance with MIL-I-24768/1.

3.3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Dimensional tolerances. Dimensions shall be as shown in the applicable specification sheet. Unless otherwise in the applicable specification sheet, the following tolerances shall apply:

a. Fractional tolerances: $\pm 1/64$ inch.

b. Decimal tolerances: ±0.005 inch.

Unless otherwise specified in the applicable specification sheet, a tolerance of $\pm 1/32$ inch is acceptable on fractional dimensions that are controlled by welding, brazing, and shock mounts. This wider tolerance shall not interfere with the interchangeability of assemblies or parts.

3.5 Preparation of metal parts. Drilling, countersinking, or tapping of metal parts shall be accomplished before plating or finish is applied.

3.6 Threads. Threads shall be right-hand, class 2 and shall be in accordance with ASME B1.1.

3.6.1 Locking devices. Locking devices shall be used to secure mechanical assemblies to provide continuous operation under conditions of heat, shock, and vibration specified herein.

3.6.2 Self-tapping devices. Self-tapping devices and sheet metal screws shall not be used in the assembly of these controls.

3.7 Welding. Welding and allied processes used in fabrication shall be in accordance with AWS WHB-1.10.

3.8 Stress relief. Metals used in fabrications and assembly shall be treated or heat-treated to prevent deterioration or failure due to stresses or other conditions resulting from working, forming, welding, and so forth, during the fabrication process. Peening after welding is an acceptable method of stress relieving in accordance with AWS WHB-1.10.

3.9 Junction boxes. The junction boxes shall be watertight and shall be in accordance with MIL-E-24142.

3.9.1 Submersible junction boxes. Submersible junction boxes larger than 60 cubic inches shall have a moisture drain on the bottom. The opening shall be closed with a ¼-inch or larger pipe plug.

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3.9.2 Junction box configuration. Junction boxes shall be so constructed that all necessary cables of required size can be brought into the box through either the top or bottom.

3.10 Limitations.

3.10.1 Mounting dimensions. Mounting dimensions shown on the specification sheets (see 3.1) shall be maintained and maximum overall dimensions shall not be exceeded.

3.10.2 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance, and strength.

3.11 Mechanical details.

3.11.1 Diagrams and identification plates.

3.11.1.1 Connection diagram. Each control shall include a connection diagram. The information shall be printed by a process that is nonfading, protected by transparent plastic, and secured to the enclosure door or cover in accordance with MIL-DTL-2036. The preferred method is to laminate the diagram between two layers of polyester film by a heat process.

3.11.1.2 Identification plate. The identification plate shall be in accordance with MIL-DTL-15024 and MIL-P-15024/5 and shall contain the following information:

a. Manufacturer.

b. Contract number.

c. Specification sheet number.

d. Date of manufacture (year).

3.11.2 Enclosure effectiveness. Unless otherwise specified (see 6.2), the degree of enclosure effectiveness shall be watertight in accordance with MIL-STD-108.

3.12 Electrical details.

3.12.1 Clearances. Minimum electrical creepage and clearances shall be in accordance with MIL-DTL-917.

3.12.2 Lugs and terminals. Lugs and terminals shall be in accordance with SAE AS7928 and MIL-T-16366.

3.12.3 Wiring space. Cabling space shall be provided for both power input cables as well as branch circuit cabling in accordance with the requirements of NFPA 70.

3.13 Painting. Painting and protection against corrosion shall be in accordance with MIL-DTL-917 or as specified (see 6.2). The exterior and interior surfaces of all enclosures shall be painted. The order of operations shall be as follows:

a. Complete all fabricating operations, such as welding, machining, drilling, tapping, and stress relieving.

b. Remove all rust and other visible corrosion products.

c. Remove all grease, oil, and dirt by solvent wiping, vapor degreasing, or caustic washing and rinsing.

d. Apply primer pretreatment coating or chemical treatment and primer.

e. Apply enamel.

3.14 Operational requirements.

3.14.1 Inclination. The controls shall operate at a rated voltage when inclined to an angle of 45 degrees from the vertical in any direction and not permanently malfunction when inclined to an angle of 60 degrees from the vertical in any direction.

3.14.2 Vibration. The controls shall withstand type I vibration tests in accordance with MIL-STD-167-1, without mechanical damage or malfunction.

3.14.3 Shock. The equipment shall withstand the grade A, class I, shock test in accordance with MIL-DTL-901, without damage or loosening of parts.

3.14.4 Corrosion. The controls shall satisfactorily withstand the salt spray test specified in 4.2.5.

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3.14.5 Noise. Unless otherwise specified (see 6.2), the noise generated by the controls shall not exceed the noise limitation for Equipment Grade C in accordance with the Shipboard Equipment Noise Requirements Appendix of MIL-STD-1474 and type III structureborne noise requirements of MIL-STD-740-2.

3.14.6 Heat. Temperature rise in current carrying parts (bus bars, connection points, fuse clips, terminals, and so forth) shall not exceed 113 °F (45 °C) above an ambient room temperature of 104 °F (40 °C). The air temperature within the controls shall not exceed 149 °F (65 °C).

3.15 Workmanship. The equipment shall be manufactured and finished in a manner that is consistent with best industry practices. Particular attention shall be paid to neatness and thoroughness of soldering, crimping, wiring, marking of parts and assemblies, plating, painting, machine screw assemblage, welding, and freedom of parts from burrs and sharp edges. See the workmanship guideline of MIL-HDBK-454 for additional information.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on the control device when a first article sample is required (see 3.2 and 6.3). This inspection shall include the examination of 4.5 and the tests of 4.2.1 through 4.2.5 and 4.6.

4.2.1 Shock. The first article sample shall be subjected to HI-shock tests in accordance with MIL-DTL-901 for grade A, class I equipment. Control components, such as switches, that have previously successfully passed the HI-shock requirements need not be retested. In lieu thereof, a simulated component may be substituted for the purpose of testing. The manner of mounting shall simulate the actual mounting on shipboard. The sample being tested shall not be reconditioned or adjusted during the testing. The specimen being tested shall be considered as failing the test in the event parts are damaged or loosened.

4.2.2 Vibration. The first article sample shall be subjected to type I vibration tests up to and including the frequency range of 33 hertz in accordance with MIL-STD-167-1. The manner of mounting shall simulate the actual mounting on shipboard. The use of resilient mounting shall not be permitted. Acceptability shall be contingent upon the ability of the equipment to withstand the specified tests and perform its principal functions during and after such tests. Failure of the equipment to function satisfactorily after testing shall be considered cause for rejection.

4.2.3 Noise. The first article sample shall be subjected to the noise level tests specified for the equipment in accordance with MIL-STD-1474 and MIL-STD-740-2 (see 3.14.5).

4.2.4 Heat test. The first article sample shall be subjected to heat testing. The heat testing shall be made under simulated conditions equivalent to normal operating conditions employing maximum rated voltage, maximum load, and rated frequency. The test shall be conducted continuously until the temperatures have remained constant for 1 hour.

4.2.5 Corrosion. The first article sample shall be subjected to salt spray tests in accordance with MIL-STD-202, method 101, condition A. The salt solution concentration shall be 20 percent.

4.2.6 Inclined operation. The control shall be tested for inclined operation at 45 degrees from the horizontal. The control shall run at rated load for a period of not less than 5 minutes at each of the following positions. At the end of the 5 minutes, the control shall be inclined to 60 degrees for 30 seconds before returning to level operation. After returning to level, the control shall be operated to demonstrate no malfunction as a result of inclined operation.

a. 45 degrees forward.

b. 45 degrees backward.

c. 45 degrees to the left side.

d. 45 degrees to the right side.

When the testing setup allows, the inclined operation test (or portions thereof) may be combined with the noise test or heat test of 4.2.3 and 4.2.4, respectively.

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4.3 Conformance inspection. Conformance inspection shall include the examination of 4.5 and the tests of 4.6.

4.4 Sampling for conformance.

4.4.1 Lot. For the purpose of sampling, a lot shall be considered to be the identical number of controls of each type as specified (see 6.2).

4.4.2 Sampling for examination and effectiveness of enclosure and dielectric test. A random sample of controls of each type shall be selected in accordance with table I from each inspection lot and shall be subjected to the examination specified in 4.5 and the tests specified in 4.6. Any control containing one or more defects and any control failing in one or more tests shall be cause for rejection.

TABLE I. Sampling for examination and effectiveness of enclosure and dielectric tests.

Number of controls in inspection lot	Number of controls in sample
2 to 10	All
11 to 15	10
16 to 25	13
26 to 40	17
41 to 65	22
66 to 110	28
111 to 180	35
181 to 300	45
301 to 500	55
501 and over	70

4.5 Examination. The controls shall be examined for compliance with the requirements specified in 3.3 through 3.13 and 3.15. Alignment of parts shall be observed to determine that components are in good working order and that the requirement for interchangeability of parts is realistic.

4.6 Tests.

4.6.1 Effectiveness of enclosure. Each first article sample and each sample selected in accordance with 4.4.2 shall be tested to determine the effectiveness of the enclosure in accordance with MIL-STD-108.

4.6.2 Dielectric. Each first article sample and each sample selected in accordance with 4.4.2 shall be subjected to dielectric testing. The sample shall be subjected for 1 minute to a dielectric test voltage, the effective potential of which is twice rated voltage plus 1,000 volts (except low voltage controls between 24 and 30 volts for which twice rated voltage plus 500 volts is applicable), applied between points of opposite polarity and between live parts and ground. The frequency of the voltage shall be 60 hertz root mean square alternating current and shall approximate a true sine wave.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The navigation light controls covered by this specification are intended for use with navigation light supply and control panels (see MIL-P-24175) on Navy ships. Current stock of navigation lighting control equipment may be used until depleted; future requisitions for navigation lighting control equipment should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Title, number, and date of applicable specification sheet.

c. Type required (see 1.2).

d. When first article inspection is required (see 3.2).

e. Materials required (see 3.3).

f. When a degree of enclosure effectiveness other than watertight may be required (see 3.11.2).

g. Painting required, if other than as specified (see 3.13).

h. Acceptable airborne noise limits, if intended application requires greater control than indicated in MIL-STD-1474 (see 3.14.5).

i. Number of controls of each type in the lot for conformance inspection (see 4.4.1).

j. Packaging requirements (see 5.1).

k. Requirements for Provisioning Technical Documentation (PTD), spare parts, and repair parts (see 6.4).

6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product that has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Provisioning. PTD, spare parts, and repair parts should be furnished as specified (see 6.2). When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.5 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.6 Subject term (key word) listing.

Pulsation
Signals

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodian:
Navy – SH

Review activity:
DLA – GS

Preparing activity:
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